

Proactive Churn Management for High-Value Subscribers

A Data-Driven Proof of Concept (PoC) Proposal for Indian Telecom Companies

Presented by

Adityabaan Tripathy

SRM Institute of Science and Technology, India

Program : GCI World 2026, Matsuo-Iwasawa Laboratory at the University of Tokyo



Navigating Our Path from Raw Historical Data to Quantifiable Revenue Protection

01

The Market Challenge & Our Business Idea

Evaluate economic and behavioral forces driving subscriber attrition, and outline our strategic business solution

02

Translating Behavior into Predictive Machine Learning Models

Examine how our predictive engine decodes raw data into clear early-warning signals for proactive intervention

03

Strategic Retention Proposals & Projected Financial ROI

Outline targeted business interventions and their direct, simulated impact on our bottom line

Shifting from Reactive Marketing to Proactive, ROI-Driven Retention



Situation

Intense price war in Indian telecom landscape with Mobile Number Portability (MNP) being completely frictionless

Data consumption is massive, yet **Average Revenue Per User (ARPU) remains severely compressed**

Market saturation forces operators to compete primarily on price, eroding operating margins

 Customers can switch providers seamlessly with zero friction



Problem

Customer Acquisition Cost (CAC) is 5-25x higher than retention costs, making churn extraordinarily expensive

Silent customer churn is **aggressively draining operating margins** without visible warning signs

Current reactive marketing blasts are too expensive, inefficient, and target the wrong subscribers

 Each lost high-value subscriber costs 5-25x more to replace



Business Idea

Deploy an early-warning Machine Learning engine to intercept flight-risk customers before they switch providers

Enable **highly targeted, ROI-positive retention campaigns** that protect revenue while minimizing waste

Transform customer retention from reactive firefighting into proactive, data-driven revenue protection

 Move from chasing lost customers to preventing departure



The Strategic Imperative: Stop reacting to churn. Start predicting and preventing it.

Utilizing Our Historical Customer Footprint to Build a Unified Customer 360 Profile

100K

Active Customer Accounts

Analyzed over a multi-month observation window to capture comprehensive behavioral patterns and lifecycle stages

 **Unified Data Architecture:** Merging behavioral and profile records creates a multi-dimensional view of each subscriber



Customer Profile Data

Captures baseline, long-term indicators

- **Account Longevity:** Tenure duration and loyalty history
- **Equipment Age:** Handset lifecycle and upgrade readiness
- **Handset Pricing:** Device value and subsidy exposure
- **Demographics:** Customer segmentation characteristics



Real-Time Behavior Data

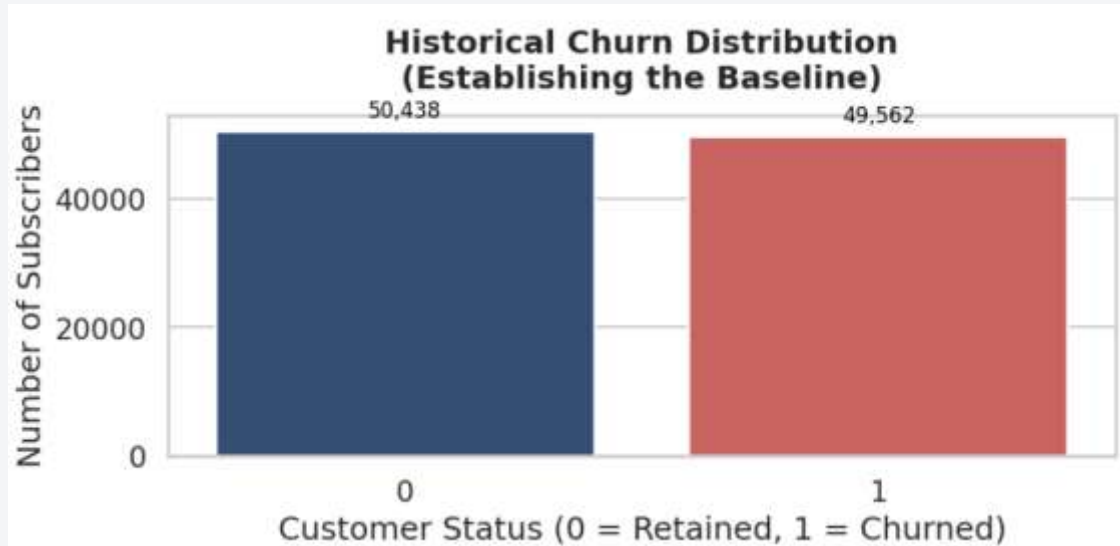
Tracks fluid, monthly metrics

- **Mean Revenue:** Monthly ARPU and billing trends
- **Minutes of Usage (MOU):** Voice and data consumption velocity
- **Overage Charges:** Bill shock indicators and plan fit
- **Network Service Issues:** Quality of experience complaints

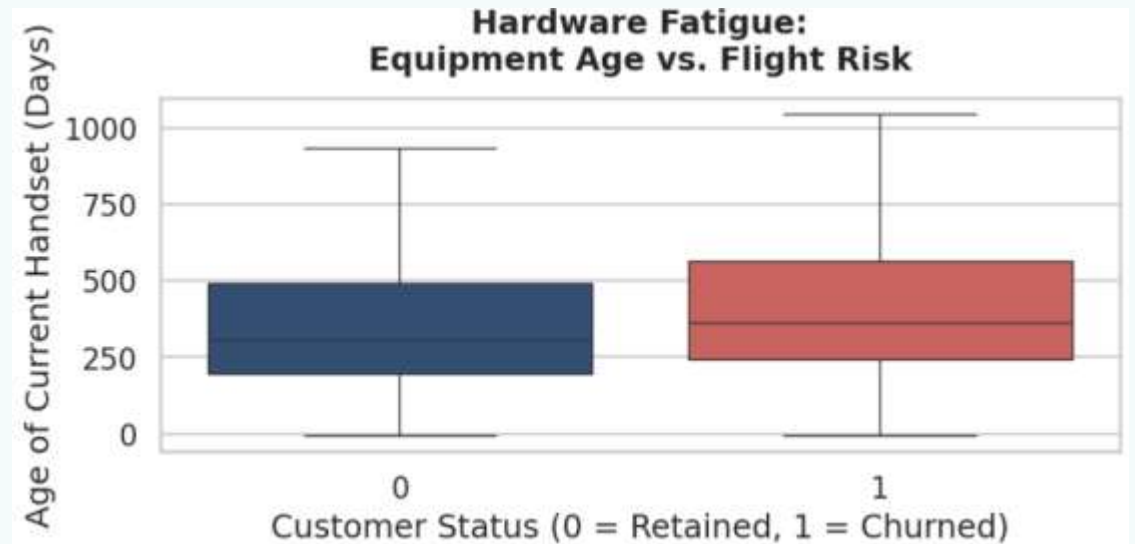
Exploratory Analysis Uncovers Visible Behavioral Predictors Long Before Churn Occurs



Historical Churn Distribution



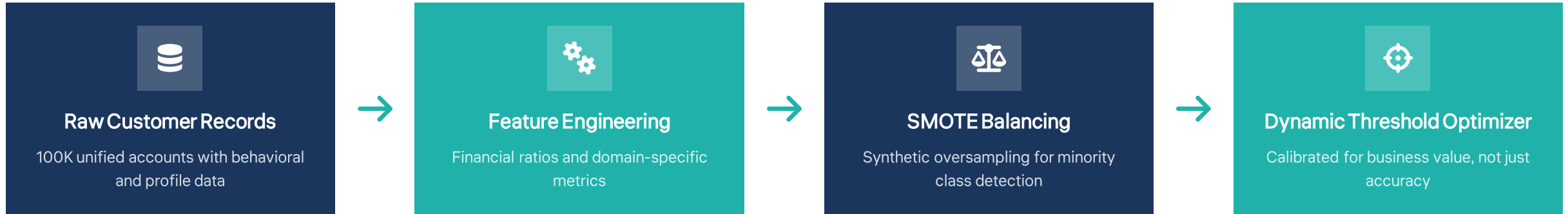
Equipment Age vs. Flight Risk



Strategic Insight

Churning customers operate on handsets that are **significantly older**, leaving them highly vulnerable to competitor upgrade promotions. This pattern reveals a clear intervention point: **proactive hardware renewal offers** can intercept flight-risk subscribers before they engage with competitor marketing.

Overcoming the Accuracy Paradox by Calibrating Our AI to Optimize for Profitability



The Accuracy Paradox

Standard models achieve **high "paper accuracy"** by predicting that most customers will stay—because most do.

This creates a **dangerous illusion of performance** while delivering zero business value, as the model never identifies actual flight risks.

The result: **wasted retention budgets** on false alarms while real churners slip away undetected.

Our Solution

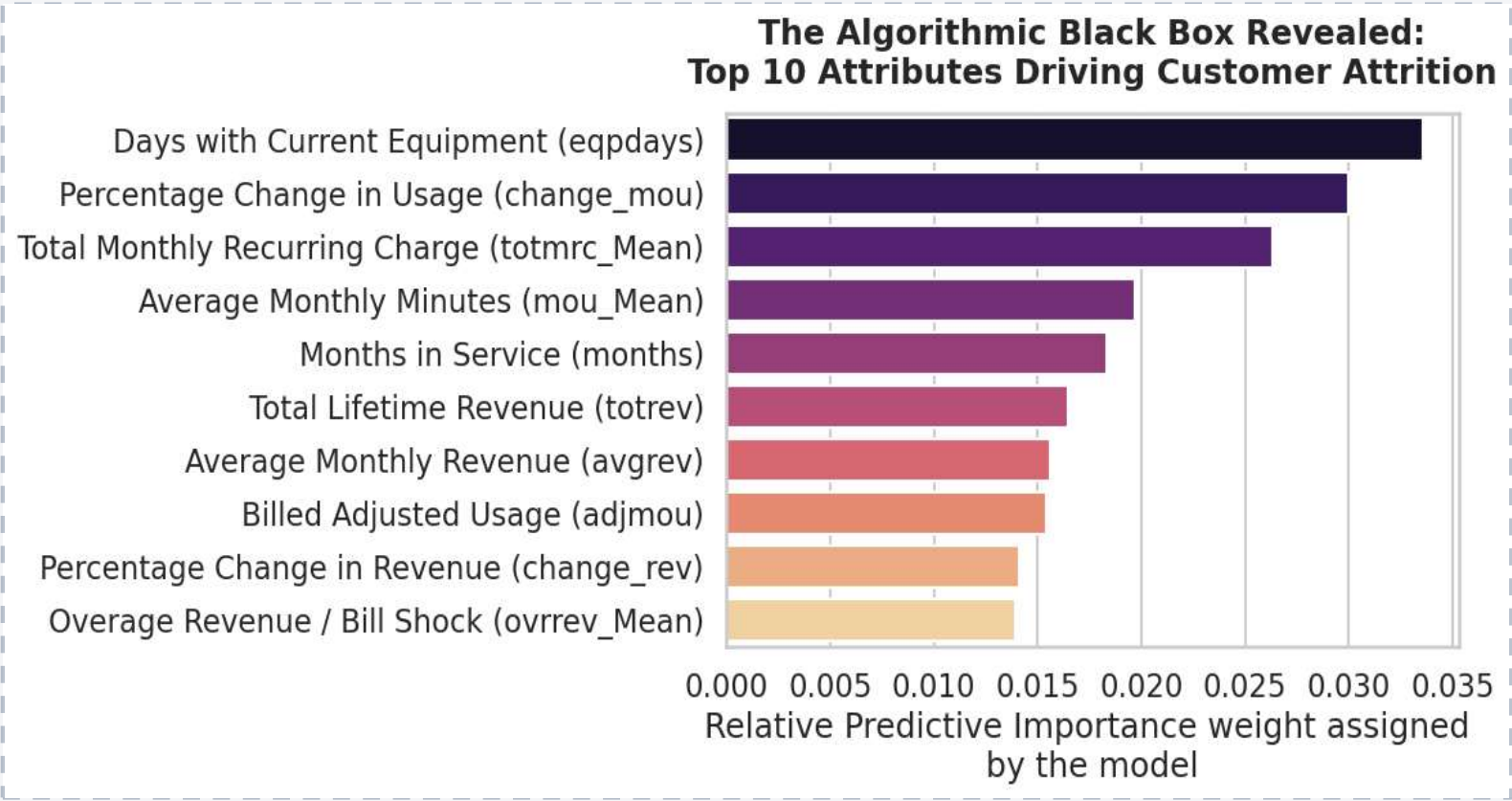
We implemented an **optimized XGBoost engine** combined with SMOTE data-balancing to force the model to pay attention to minority churn signals.

The **Dynamic Threshold Optimizer** scans the entire risk spectrum, maximizing true risk identification rather than playing it safe.

This ensures **budget is spent only on subscribers genuinely preparing to leave**, not on false alarms.

The AI Identifies Equipment Age and Recent Bill Shock as the Primary Triggers of Attrition

Top 10 Attributes Driving Customer Attrition



Algorithm Insights

The algorithm **bypasses** human intuition, analyzing complex behavioral interactions invisible to traditional analysis.

It highlights that **long equipment lifecycles** combined with **sudden drops in monthly usage velocity** are the ultimate signals of an inactive subscriber.

Engineered metrics like **"Cost Per Minute"** also rank in the top tier, revealing plan misalignment.

Critical Finding

When a user's phone gets **old** and their usage **drops**, they are silently preparing to exit. These are **actionable, early-warning signals**.

Deploying Automated, Precision Interventions to the Highest-Risk Customer Cohorts



Strategic Pillar 1

Targeted Hardware Renewals



Action

Isolate subscribers in the highest-risk decile who have equipment lifecycles exceeding 12 months. These customers show the strongest correlation between hardware age and flight probability.



Incentive

Push a proactive, personalized app notification offering an attractive, subsidized handset upgrade paired with a contract extension. Transform tech fatigue into loyalty.

Expected Impact: 15-25% reduction in hardware-driven churn



Strategic Pillar 2

Bill Shock & Plan Rightsizing



Action

Identify high-value users whose risk scores are driven by high overage charges and sudden drops in voice/data usage. These subscribers are experiencing bill shock and plan misalignment.



Incentive

Have customer care proactively reach out to offer right-sized, unlimited tier upgrades, waiving current overages to restore brand loyalty and prevent silent departure.

Expected Impact: 20-30% reduction in bill-shock-driven churn

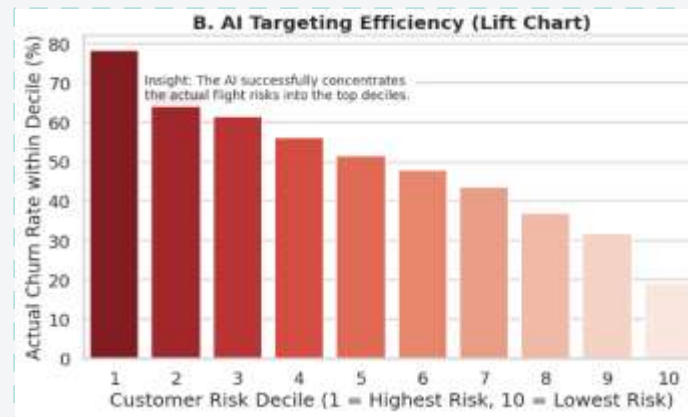
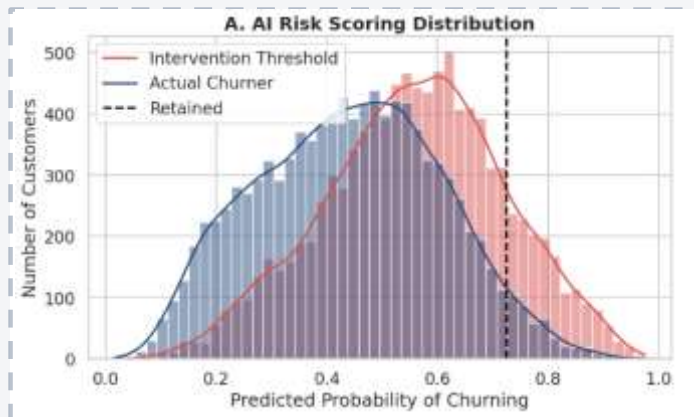


Automated Execution: Both pillars are triggered automatically by the AI engine when risk thresholds are breached, ensuring zero-delay intervention

Predictive Targeting Restricts Spend to Vulnerable Tiers, Maximizing Financial ROI



Performance Dashboard



\$18,400.00

Simulated Campaign Net Profit

High-value retention campaigns driven by this AI engine yield an optimized return on investment, converting a historical cost center into a direct revenue protector.



Financial Logic

- Focus spend on top 10% riskiest users
- Avoid wasting capital on loyal customers
- Gross revenue protected outweighs incentive costs
- Massive net profit secured for operations

Translating Complex Data into a Predictive and Profitable Business Engine



Diagnosing the Root Cause

We moved beyond surface-level metrics to **pinpoint hardware fatigue and sudden usage drops** as the primary behavioral triggers for churn. Our exploratory analysis revealed that customers operating on older handsets with declining usage velocity are silently preparing to exit—providing clear, actionable early-warning signals.



Empowering the Algorithm

By engineering **domain-specific financial signals** and utilizing SMOTE to balance the data, we trained an XGBoost model optimized specifically for the complexities of human behavior. This approach overcomes the accuracy paradox, ensuring the model identifies true flight risks rather than playing it safe with statistically correct but commercially useless predictions.



Calibrating for Profit

Instead of accepting standard "safe" accuracy, we **dynamically optimized the decision threshold** to actively isolate the top 10% of high-value flight risks. This calibration protects millions in revenue while minimizing campaign costs, transforming customer retention from a reactive expense into a proactive profit center.



The Bottom Line

This Proof of Concept demonstrates that predictive churn management is not a technical exercise—it is a **strategic imperative** for protecting operating margins in a saturated, price-compressed market.

Industry Research & Academic References

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S. Wu, “Customer Churn Prediction in Telecom Based on Machine Learning,” Highlights in Science, Engineering and Technology, vol. 94, pp. 113–118, Apr. 2024, doi: 10.54097/snc09915.



Methodology Validation: The methodologies applied in this proposal are heavily supported by recent 2024 and 2025 industry research, highlighting XGBoost and advanced Machine Learning techniques as the gold standard for modern telecommunications churn prediction.

Image References

1

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All the other graphs that have been used in this presentation have been taken directly from the output generated in the colab notebook which has been submitted.